Stat 17 Week 4

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Probability Topics — Part 2: Events & Probability Rules

- **Events**
  - An event is any collection of outcomes from the sample space
  - Impossible events have probability 0
  - Certain events have probability 1
  - Events with very small probability (often < 0.05) are considered unusual
- **Basic Probability Rules**
  - $0 \leq P(A) \leq 1$ for any event A
  - The probabilities of all outcomes in the sample space sum to 1
- **Complements**
  - The complement of event A, denoted A^c, includes all outcomes not in A
  - Complement rule:
    $$P(A^c) = 1 - P(A)$$
- **Unions and Intersections**
  - Intersection: $A \cap B$ (both A and B occur)
  - Union: $A \cup B$ (A or B or both occur)
- **Mutually Exclusive Events**
  - Two events are mutually exclusive if they share no outcomes
  - If A and B are mutually exclusive:
    $$P(A \cup B) = P(A) + P(B)$$
- **Addition Rule (OR)**
  - For any two events A and B,
    $$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$
  - If A and B are **mutually exclusive** (cannot occur together, i.e. $P(A \cap B) = 0$), then
    $$P(A \cup B) = P(A) + P(B).$$

- **Decomposition of an Event** Any event A can be written as the union of two disjoint events:

$$A = (A \cap B) \cup (A \cap B^c).$$

Since $(A \cap B)$ and $(A \cap B^c)$ are mutually exclusive,

$$P(A) = P(A \cap B) + P(A \cap B^c).$$

- **Conditional Probability**
  - The probability of A given B is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

  - The "given" event always appears in the denominator

- **Multiplication Rule(AND)**
  - For any two events A and B,

$$P(A \cap B) = P(A \mid B)\, P(B).$$

  If A and B are **independent**, then

$$P(A \cap B) = P(A)\, P(B).$$

- **Independent Events**
  - Events A and B are independent if one does not affect the probability of the other
  - Equivalent conditions:
    * $P(A|B) = P(A)$
    * $P(B|A) = P(B)$
    * $P(A \cap B) = P(A)P(B)$

- **At Least One Events**
  - These are often easiest to compute using complements:

$$P(\textit{at least one}) = 1 - P(\textit{none})$$

- **Law of Total Probability:** If $\{B_1, B_2, \ldots, B_n\}$ is a partition of the sample space with $P(B_i) > 0$, then

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)\, P(B_i).$$

- **Tree Diagrams and Bayes' Rule**
  - Tree diagrams help visualize conditional and joint probabilities
  - Bayes' Rule allows probability updates given new information